Li Falah-Jurnal Studi Ekonomi Dan Bisnis Islam
Volume 10 (No.1 2025) 85-96
P-ISSN: 2541-6545, E-ISSN: 2549-6085

Li Falah
Journal of Islamic Economics and Business Studies
journal homepage: http://ejournal.iainkendari.ac.id/lifalah

Seasonal ARIMA Forecasting of MSME Sales: A Case Study on KleponHolic

Muh. Syariel Nizam[1], Muhammad Rais Rahmat Razak[2], Andi Riska Andreani Syafaruddin[3], Pratiwi Ramlan[4]

[1,3,4] Faculty of Economics and Business, Muhammadiyah University of Sidenreng Rappang, Indonesia
[2] Master of Public Administration, Muhammadiyah University of Sidenreng Rappang, Indonesia
*e-cmail: *syariel22@gmail.com, [2] mraisrahmat@gmail.com, [3]riska.andreani93@gmail.com, [4]pratiwiramlan@yahoo.com*

ARTICLE INFO

Article History:
Received 17 April
1st Received in revised form 19 Mei 2025
2 st Received in revised form 10 June 2025
Available online 30 Juni 2025

Keywords:
Forecasting; Seasonal ARIMA; Sales Forecasting.

JEL classification:
E17, C22, M31

DOI:
https://doi.org/10.31332/lifalah.v10i1.12322

A B S T R A C T

This study analyzes sales forecasting using the Seasonal ARIMA method on KleponHolic, a culinary MSME in Sidenreng Rappang Regency. The objectives are to identify seasonal patterns in KleponHolic's sales data, evaluate the accuracy of the Seasonal ARIMA model, and provide strategic recommendations for stock management and data-driven marketing. This research applies a descriptive quantitative approach with Univariate Time Series Analysis, using monthly sales data from 2021 to 2024 collected through observation, documentation, and informant validation. The data were processed through seven stages of Seasonal ARIMA modeling, including data exploration, stationarity testing, model identification, parameter estimation, diagnostics, forecasting, and interpretation. Unlike previous studies that mostly applied Seasonal ARIMA in large-scale sectors such as climatology, transportation, and natural resource management, this research addresses a gap by applying the method to the MSME food and beverage sector, where studies remain limited. The findings not only contribute to practical business strategies for KleponHolic but also provide theoretical implications by expanding the application of Seasonal ARIMA in MSME contexts, highlighting how seasonal demand patterns influence consumption behavior and business decision-making.

1. Introduction

Forecasting is one of the fundamental components in business decision-making, particularly in planning strategies to address ever-changing market dynamics. In business contexts, forecasting is widely applied to anticipate demand fluctuations, optimize inventory, and minimize operational risks. Among various forecasting techniques, the Seasonal Autoregressive Integrated

Moving Average (Seasonal ARIMA) has been extensively used due to its ability to capture seasonal variations in time series data. Moreover, the integration of Information and Communication Technology (ICT) has been proven to transform organizational structures and decision-making processes, making them more efficient, transparent, and adaptive to rapid changes (Razak et al., 2021).

Several previous studies have demonstrated the application of Seasonal ARIMA in different sectors. For instance, (Kartono et al., 2020) applied it to predict water availability in a reservoir in Pati, Indonesia. At the international level, the method has been employed for forecasting rainfall (Fatima et al., 2022), energy demand (Khan et al., 2021), and tourist arrivals in nepal (Paudel et al., 2024). However, most of these studies concentrated on macro-level sectors such as climatology, transportation, and natural resource management. There is still a clear research gap in applying Seasonal ARIMA within the Micro, Small, and Medium Enterprises (MSMEs) sector, particularly in the Food and Beverage (F&B) industry, despite its significant contribution to both local and global economies.

As highlighted in prior research, forecasting plays a crucial role in reducing uncertainty and helping businesses prepare for future market fluctuations (Syafaruddin et al., 2024). In the context of MSMEs, these challenges are even more pronounced due to limited resources. SMEs often lack the resources and technical expertise required for implementing advanced forecasting models, making it challenging to manage demand fluctuations effectively (Gaertner et al., 2024).

In Indonesia, the MSME sector contributes 61% to the national Gross Domestic Product

Data UMKM 2018-2023

Tahun	2018	2019	2020	2021	2022	2023
Jumlah UMKM (Juta)	64.19	65.47	64	65.46	65	66
Pertumbuhan (%)		1.98%	-2.24%	2.28%	-0,70%	1,52%

Figure 1. MSME Data for the Years 2018–2023
Source: Kadin Indonesia (2023)

(GDP), equivalent to IDR 9,580 trillion, and absorbs 97% of total employment (Coordinating Ministry for Economic Affairs, 2023). MSMEs play an important role in the national economy, yet they often face various obstacles such as limited capital, low-quality human resources, and underutilization of technology (Ramlan et al., 2025). To remain competitive in the era of globalization, MSMEs are required to be more adaptive by implementing data-based strategies and utilizing information technology. In this context, KleponHolic an emerging culinary MSME in Sidenreng Rappang Regency faces critical challenges related to seasonal demand fluctuations, resulting in stock imbalances and potential profit losses during peak seasons. This condition highlights the urgency of adopting data-driven forecasting models that are suitable for the MSME sector.

This study provides three main contributions. Theoretically, it expands the literature by applying Seasonal ARIMA to MSME sales forecasting, bridging the gap between large-scale sectoral studies and small-scale business applications. Practically, it offers strategic insights for KleponHolic to optimize inventory management and marketing strategies through accurate sales forecasting. Academically, it contributes to the global discourse on time series analysis by showcasing how predictive models can be adapted to MSMEs, which represent the backbone of many national economies. The novelty of this research lies in positioning Seasonal ARIMA as a reliable forecasting tool for MSMEs in the F&B sector, thereby extending its application beyond traditional macro-level domains.

In Indonesia, the Seasonal ARIMA method has been applied to predict water availability in a reservoir in Pati, highlighting its usefulness not only in capturing seasonal patterns but also in generating relevant predictive data for natural resource management. Nevertheless, most previous studies have concentrated on macro-level sectors such as transportation, climatology, and large-scale natural resource management, with limited research exploring its application in the Micro, Small, and Medium Enterprises (MSMEs) sector—particularly in Food and Beverage (F&B) business forecasting. The novelty of this research, therefore, lies in applying the Seasonal ARIMA model within the MSME F&B context, offering a distinct perspective from prior macro-focused studies. In line with advancements in technology and data science, predictive approaches have become an essential tool in supporting data-driven business decision-making processes, including within the MSME sector. This is especially relevant in the Indonesian economy, where MSMEs play a crucial role as the backbone of national growth, contributing significantly to GDP and serving as a primary source of employment for the majority of the workforce (Maulifah et al., 2024).

2. Literature Review

2.1 Forecasting in Business Context

Forecasting is the process of predicting future events or trends based on historical data. In the business context, forecasting plays a critical role in anticipating demand, optimizing inventory, and supporting strategic decision-making (Kanatani et al., 2019). According to (Kusumawardani et al., 2019), forecasting involves estimating the quantity, quality, timing, and location of particular needs to meet predetermined targets. (Lestari & Yanti, 2023) emphasize that forecasting helps anticipate future trends, enabling organizations to make better-informed decisions.

Globally, forecasting is increasingly recognized as a key tool for small and medium-sized enterprises (SMEs) to remain competitive in uncertain markets. Accurate forecasting improves SMEs' ability to manage resources, reduce uncertainty, and adapt to rapid changes (Madrid et al., 2016). It has also been linked to enhanced financial performance and operational resilience, particularly in dynamic industries such as food and beverage (Makridakis et al., 2020).

Forecasting methods are broadly classified into qualitative and quantitative approaches (Makridakis et al., 2020). While qualitative approaches rely on expert judgment, quantitative methods—such as time series forecasting—use historical data to uncover trends and patterns. This study adopts a quantitative approach using univariate time series data from KleponHolic.

2.2 Seasonal Factors in Sales

Seasonal factors represent recurring fluctuation patterns in data, occurring either monthly or annually. (Lusiana & Yuliarty, 2020) state that seasonal factors significantly influence consumer behavior, particularly during holidays or special celebration periods. In culinary businesses such as KleponHolic, seasonal moments like Eid al-Fitr, school holidays, and year-end festivities often lead to spikes in demand. International studies report similar patterns. For instance, (Jiang et al., 2020) highlight that European SMEs in retail and hospitality experience demand surges during Christmas and summer holidays.

While increased demand during specific periods can boost profits, failure to anticipate it properly may result in stock imbalances and financial losses. Therefore, understanding and analyzing seasonal patterns is crucial for effective marketing strategy planning and inventory management. As noted by (Huber & Stuckenschmidt, 2020), accurately forecasting demand during special calendar days is essential for retailers to make informed production and ordering decisions, thereby reducing the risk of overproduction or underproduction during peak seasons.

2.3 Forecasting Methods for SMEs

Beyond SARIMA, SMEs apply various forecasting methods to address demand variability. Traditional models like exponential smoothing and ARIMA are still widely used for their interpretability and low computational requirements. However, recent research demonstrates

that advanced methods such as machine learning (ML) and deep learning provide better accuracy when handling complex, non-linear datasets (Falatouri et al., 2022).

Nevertheless, SMEs often face resource limitations. As (Maryam & Goran, 2020) points out, the challenge lies not only in accuracy but also in the feasibility of implementation and alignment with managerial decision-making. Thus, models like SARIMA remain practical for SMEs seeking a balance between technical accuracy and business applicability.

2.4 Seasonal ARIMA Method

Seasonal ARIMA (SARIMA) extends ARIMA by integrating seasonal components into time series analysis. It is denoted as SARIMA (p,d,q)(P,D,Q)[s], where p, d, q represent non-seasonal parameters, P, D, Q represent seasonal parameters, and s indicates the seasonal cycle length. According to (Ridla et al., 2023), SARIMA is highly effective for data with consistent seasonal patterns, as it captures both short- and long-term fluctuations. For KleponHolic, which shows strong seasonal sales, SARIMA is an appropriate model because it offers accuracy while remaining feasible for SMEs.

2.5 Forecast Accuracy Evaluation

To evaluate the accuracy level of the forecasting model, several error measures are used, including Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The interpretation of MAPE values is generally categorized as follows: a MAPE value of less than 10% indicates a highly accurate model; 10–20% is considered good; 20–50% is acceptable; and greater than 50% is considered inaccurate. This evaluation helps assess how well the model predicts future data and supports the decision-making process by determining the model's overall reliability and feasibility.

2.6 Previous Studies

Several previous studies have shown that the Seasonal ARIMA method has been widely applied for forecasting across various sectors. For example, (Gunawan & Ramadani, 2023) compared the effectiveness of Seasonal ARIMA and Prophet in predicting inventory levels at PT XYZ. The results revealed that Seasonal ARIMA with the model (0, 0, 0)(0, 1, 1, 12) produced more accurate forecasts, with an RMSE value of 3.61 compared to Prophet's 4.02. Seasonal ARIMA's strength lies in its ability to capture seasonal patterns more effectively, thereby supporting more efficient inventory management for the company.

(Hassyddiqy & Hasdiana, 2023) employed the ARIMA model to forecast clothing sales at Huebee Indonesia. The ARIMA (0, 1, 1) model yielded highly accurate forecasting results with a MAPE of 8.58%, which proved useful for production planning and distribution strategies.

In another study, (Septiana & Bangun, 2023) compared the performance of Seasonal ARIMA and the Feed Forward Neural Network (FFNN) method in forecasting seasonal milk production. Seasonal ARIMA outperformed FFNN with a MAPE of 1.775% compared to 7.5226%, demonstrating Seasonal ARIMA's effectiveness in handling seasonal time series data.

However, the application of the Seasonal ARIMA method in the context of micro, small, and medium enterprises (MSMEs), particularly in the Food and Beverage (F&B) sector, remains very limited. Therefore, this study aims to fill that gap by implementing the Seasonal ARIMA method to analyze seasonal sales patterns in the KleponHolic MSME. It is expected that the results of this research will provide practical contributions for MSME actors in managing stock strategies and marketing efforts based on predictive data.

2.6 Conceptual Framework

This research begins with the identification of a problem in the form of fluctuating sales at KleponHolic that have not been systematically documented. Subsequently, secondary data were collected in the form of monthly sales data over 48 months (2021 2024). The data were analyzed using the Seasonal ARIMA approach, which involves identifying seasonal patterns, conducting stationarity tests, modeling, validating accuracy (MAPE, MAE, RMSE), and forecasting sales for the year 2025. The forecasting results were then validated through informant statements from the KleponHolic business owner, in order to assess the alignment between the model's output and real-world conditions, as well as to evaluate external factors not captured by numerical data.

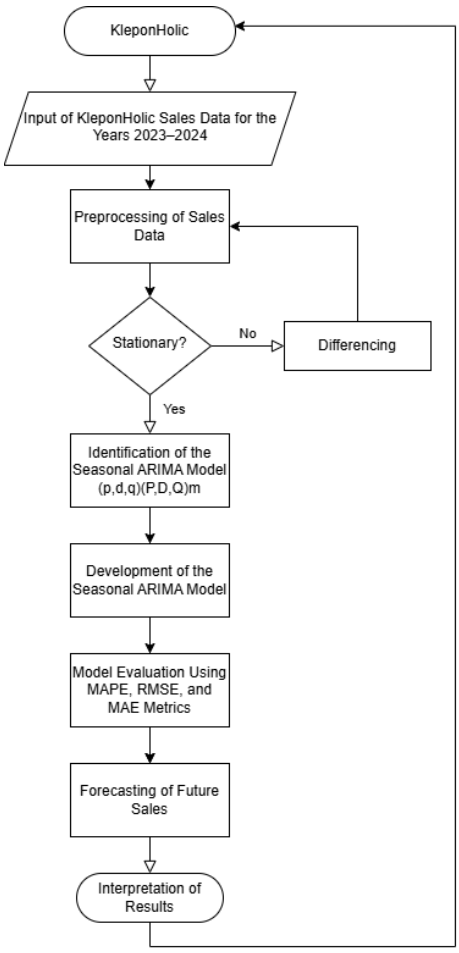

Figure 2. Conceptual Framework
Source: processed by Author (2025)

3. Research Method

3.1 Research Approach and Type

This study employs a quantitative approach. Quantitative research is an investigation of social phenomena based on the testing of a theory composed of variables that are measured numerically and analyzed using statistical procedures to determine whether the predictive generalizations of the theory are accurate (Ali et al., 2022). The type of research used in this study is descriptive quantitative. According to (Razak & Harfiah, 2018), descriptive quantitative research presents facts and explains the actual conditions of the research object as they are, and attempts to analyze the data to validate their truth based on field findings. Furthermore, (Berlianti et al., 2024) explain that descriptive quantitative research is a method aimed at describing or illustrating a phenomenon or condition occurring in a specific population using numerical data.

3.2 Research Location and Time

This research was conducted at the KleponHolic outlet located on Jalan Angkatan 66, Rappang Sub-district, Panca Rijang District, Sidenreng Rappang Regency, South Sulawesi. The study was carried out over a two-month period, specifically in March and April 2025.

3.3 Population and Sample (Research Target)

In a univariate time series study such as this, sales data serves as the primary focus of analysis, and thus the conventional concept of population and sampling is not applied. However, to support the interpretation of results and gain deeper contextual insights regarding seasonal factors affecting sales, the researcher involved a key informant—namely, the owner of KleponHolic. This informant was selected purposively, based on the consideration that they possess the most comprehensive understanding of the business operations, sales trends, and strategies that have been implemented.

3.4 Data Collection Techniques and Instrument Development

The data collection techniques in this study consist of three main methods. The first is a literature review, which was conducted to obtain a strong theoretical foundation regarding forecasting methods, seasonal patterns, and the implementation of SARIMA in the context of business and sales management.

The second is documentation, which was used to obtain historical sales data of KleponHolic covering the period from January 2021 to December 2024. This data was gathered from the business's internal records, which have been documented regularly on a monthly basis.

The third method is key informant information gathering, in which direct information was requested from the owner of KleponHolic regarding market trends, seasonal factors affecting sales, and responses to the forecasting results. This information was not analyzed in-depth qualitatively, but rather used as interpretive support for the quantitative outcomes generated by the forecasting model.

3.5 Instruments and Materials Used

In this study, software tools were used to support the data processing and analysis procedures. The primary software utilized was RStudio version 2024.12.1 Build 563, which served for time series data processing and statistical analysis. Additionally, Microsoft Excel was employed in the initial stage for organizing and cleaning the sales data. Further analysis was supported by the use of Python programming language, which was executed within the RStudio environment to run time series analysis scripts more flexibly.

The material used in this research consisted of monthly sales data of KleponHolic for the period 2021 to 2024, expressed in Indonesian Rupiah, obtained directly from the business's internal financial records.

3.6 Data Analysis Technique

The data collected were analyzed using the Seasonal ARIMA (SARIMA) method. The analysis process began with an initial exploration of the data to identify seasonal patterns and general trends. Next, a stationarity test was conducted using the Augmented Dickey-Fuller (ADF) Test to ensure that the data met the fundamental assumptions of the SARIMA model. Subsequently, the researcher determined the optimal model parameter combination—(p,d,q)(P,D,Q)[s]—based on the ACF and PACF plots, as well as evaluations of the AIC and BIC values. The selected model was then further evaluated by testing the residuals using the Ljung-Box test to confirm the absence of remaining autocorrelation. Model accuracy was also assessed by calculating MAPE (Mean Absolute Percentage Error), MAE (Mean Absolute Error), and RMSE (Root Mean Square Error). These values help measure the model's precision in forecasting future sales data.

The justification for using SARIMA over other forecasting methods, such as Prophet or machine learning-based approaches, lies in the characteristics of the dataset and research context. First, the sales data consist of monthly observations with clear seasonal patterns and a relatively limited time span (2021–2024), which makes SARIMA particularly suitable for capturing both trend and seasonality effectively. Second, SARIMA offers greater interpretability compared to black-box machine learning models, which is essential in the context of small and medium-sized

enterprises (SMEs) such as KleponHolic, where decision-makers may prefer transparent methods. Finally, compared to Prophet, SARIMA provides more rigorous statistical diagnostics (e.g., ADF, ACF, PACF, Ljung-Box), ensuring that the model assumptions are systematically tested.

The final step in the analysis involved forecasting sales for the year 2025 and comparing the forecasting results with insights obtained from the business owner. While the business owner's input was valuable for contextual interpretation, this reliance should be acknowledged as a limitation, as it does not provide an independent or external form of validation. Future research is encouraged to incorporate alternative validation strategies, such as cross-validation or benchmarking against other forecasting models, to enhance robustness.

4. Result

This study aims to analyze the trends and seasonal patterns in the sales of KleponHolic products and to generate accurate sales forecasts for the year 2025 using the Seasonal ARIMA method. The research has two main focuses: first, to develop a time series model capable of capturing fluctuations in historical data, and second, to formulate business implications based on the forecasting results within the operational context of culinary MSMEs. The following is a comprehensive discussion based on the results of the data analysis.

4.1 Historical Patterns and Sales Trends

Based on KleponHolic's sales data from January 2021 to December 2024, a fairly consistent seasonal pattern was identified, in which sales tend to increase significantly in June, August, and December, while February and January show a declining trend.

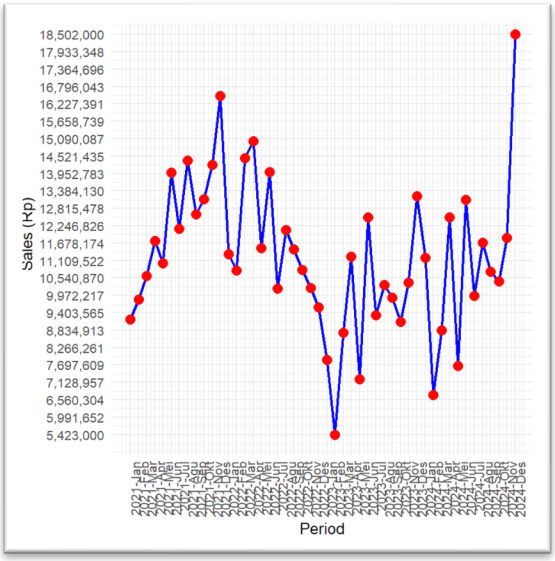

Figure 3. KleponHolic Sales Trend Chart for the Period 2021–2024

Source: Researcher (2025), processed using RStudio 2024.12.1 Build 563.

Based on KleponHolic's sales data from January 2021 to December 2024, a consistent seasonal pattern was identified, with significant increases in June, July, and December, while January and February tended to show declines. These fluctuations align with cultural and economic cycles in Indonesia: June and July coincide with Eid al-Fitr celebrations and school holidays, whereas December aligns with Christmas and New Year festivities, both of which typically boost food consumption. In contrast, the decline in January and February may be attributed to decreased consumer purchasing power following high expenditures at the end of the previous year.

From a business perspective, this highlights the need for MSMEs to anticipate demand cycles. Strategically, KleponHolic can leverage these insights by strengthening inventory and production capacity ahead of festive seasons and introducing promotional campaigns during off-peak months.

More broadly, these findings confirm the critical role of seasonal demand in shaping sales performance for culinary MSMEs, echoing previous evidence that seasonal fluctuations strongly influence retail and F&B sectors (Retnaning et al., 2021).

4.2 Stationarity and Data Transformation

The Augmented Dickey-Fuller (ADF) test on the original data indicated non-stationarity (p-value > 0.05). To address this, two non-seasonal differencing operations (d = 2) and one seasonal differencing (D = 1) with a seasonal period of s = 12 were applied. The transformed data showed a p-value < 0.05, confirming stationarity. Ensuring data stationarity is critical because non-stationary data can bias parameter estimation and reduce predictive accuracy, which is particularly detrimental when forecasting for small enterprises where decision-making relies on precise projections.

4.3 Model Identification and Estimation

The model identification process was carried out using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, along with automatic optimization through the auto.arima() function in RStudio. Based on the analysis results, the best-fitting model was identified as Seasonal ARIMA (0,2,1)(0,1,0)[12]. This model was selected due to its lowest values of Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), as well as residuals that displayed a random pattern with no significant autocorrelation according to the Ljung-Box test, indicating that the model left no unexplained structure (white noise).

Structurally, the model includes a non-seasonal Moving Average (MA) component of order 1, which serves to capture error fluctuations from previous periods, thereby improving future predictions. Two rounds of non-seasonal differencing (d = 2) were applied to eliminate the strong linear trend in the data, making it stationary in mean. In addition, one seasonal differencing (D = 1) was used to address the annually recurring pattern identified in the data, with a seasonal cycle length of 12 months (s = 12). Interestingly, the model does not include any Autoregressive (AR) components or additional seasonal parameters (P and Q), suggesting that the influence of past values was not strong enough to be included in the model. Thus, this simpler configuration turned out to produce the most optimal result.

The advantage of this model lies in its balance between structural simplicity and forecasting accuracy. Although it contains only a single core component—MA(1)—the model effectively captures the sales dynamics of KleponHolic. In the context of micro, small, and medium enterprises (MSMEs) like KleponHolic, a model that is both simple and accurate is highly beneficial, as it is easier to understand and implement by business owners who may not have a background in statistics or information technology. Therefore, the Seasonal ARIMA (0,2,1)(0,1,0)[12] model is not only technically valid but also practically relevant for data-driven forecasting and business decision-making.

4.4 Model Accuracy Evaluation

The evaluation was carried out through residual analysis and statistical testing using the Ljung-Box test, as well as by calculating several error metrics, namely MAPE, MAE, and RMSE. The results of the evaluation are presented below.

Table 1. Results of Ljung-Box Test and MAPE, MAE, and RMSE Metrics

Type of Test	Value	Interpretation
Uji Ljung-Box (p-value)	0.6897	There is no significant autocorrelation in the residuals, and the residuals are random, indicating that the model passes the diagnostic test.
MAPE	9,13%	Highly accurate because the value is less than 10%

MAE	Rp953.000	The average prediction deviates by approximately ±Rp953 thousand from the actual value.
RMSE	Rp1.501.242	The average prediction error is approximately ±Rp1.5 million, giving greater weight to larger errors.

These results demonstrate that SARIMA is reliable in capturing KleponHolic's sales patterns. The MAPE value of 9.13% aligns with the findings of (Gunawan & Ramadani, 2023), who reported SARIMA's superiority over Prophet in handling seasonal inventory forecasts. Although (Septiana & Bangun, 2023) achieved a lower MAPE of 1.77% in dairy production forecasting, the higher error rate in this study may stem from external factors influencing MSMEs in the culinary sector, such as fluctuating consumer spending power and local market dynamics.

4.5 Forecasting Results

Based on the Seasonal ARIMA (0,2,1)(0,1,0)[12] model used, the sales forecast for KleponHolic in 2025 shows a fluctuating pattern with a significant increase in the middle to the end of the year. The forecasting results are presented below.

Table 2. Sales Forecasting Results of KleponHolic for the Year 2025

Month	Point Forecast	Lo 80%	Hi 80%	Lo 95%	Hi 95%
January 2025	16,843,364.14	14,522,038.45	19,164,689.83	13,293,202.86	20,393,525.42
February 2025	12,695,728.27	9,295,074.71	16,096,381.84	7,494,877.46	17,896,579.09
March 2025	15,147,092.41	10,836,619.51	19,457,565.31	8,554,792.99	21,739,391.83
April 2025	19,194,456.55	14,047,782.82	24,341,130.28	11,323,298.32	27,065,614.77
May 2025	14,669,820.69	8,724,976.89	20,614,664.48	5,577,966.69	23,761,674.69
June 2025	20,455,184.82	13,732,727.69	27,177,641.96	10,174,073.84	30,736,295.81
July 2025	17,657,548.96	10,168,120.49	25,146,977.43	6,203,456.55	29,111,641.37
August 2025	19,736,913.10	11,485,004.49	27,988,821.70	7,116,707.94	32,357,118.25
September 2025	19,131,277.23	10,117,326.81	28,145,227.66	5,345,629.70	32,916,924.77
October 2025	19,149,641.37	9,371,299.99	28,927,982.74	4,194,958.75	34,104,323.99
November 2025	20,919,005.51	10,371,945.70	31,466,065.32	4,788,669.50	37,049,341.52
December 2025	27,911,369.65	16,589,826.66	39,232,912.63	10,596,563.83	45,226,175.46

The 2025 sales forecast using the SARIMA (0,2,1)(0,1,0)[12] model reveals that KleponHolic's sales are projected to rise sharply towards the middle and end of the year. December is expected to record the highest sales, reaching approximately Rp27.91 million (see Table 2). The prediction intervals at the 80% and 95% confidence levels indicate higher uncertainty in the early months, particularly January and February, suggesting potential risks of declining sales. These findings imply that KleponHolic must increase its production and distribution capacity during peak months while strengthening promotional strategies in the early months to maintain stable cash flow. From an academic perspective, this result is consistent with Falatouri et al. (2022), who reported that SARIMA outperforms LSTM when dealing with strongly seasonal data.

Based on the forecast results, it can be observed that sales are expected to fluctuate throughout the year, with a tendency to increase during the middle to the end of the year. The highest projected sales point is expected to occur in December, with an estimated value of approximately Rp27.91 million. The prediction intervals at 80% and 95% confidence levels provide insights into the degree

of monthly uncertainty, where early months—such as January and February—indicate a potential decline that should be addressed with specific strategies. Understanding this pattern enables business owners to plan production and promotions more effectively in response to seasonal dynamics.

Insights from the KleponHolic business owner further support these forecasting results. Factors such as the use of premium ingredients, attractive packaging, unique flavor variations, and affordable pricing were identified as key drivers of consumer appeal. The business owner also acknowledged that the forecasting model is fairly accurate and useful as a foundation for decision-making, especially in managing inventory and designing promotional strategies ahead of peak sales periods such as June, July, and December. However, it should be noted that certain external factors—such as changes in sales location, operational hours, and daily promotional activities—were not captured in the historical data. Therefore, more comprehensive data collection is necessary to generate forecasts that are more accurate and adaptive to market dynamics in the future.

Based on the analysis and informant input, it is recommended that KleponHolic strengthen its promotional strategies during the low season (particularly January and February) through discount programs, bundling offers, or customer loyalty initiatives. On the other hand, during high-sales months, the business should expand production capacity and distribution logistics to avoid stock shortages. Marketing strategies should also highlight the product's core strengths through visual branding and social media engagement, including leveraging customer testimonials and collaborating with local influencers. This data-driven approach is highly recommended for other MSMEs in Sidenreng Rappang Regency to enhance operational efficiency, planning, and sustainability in response to evolving market behavior.

5. Implication

The practical implication of this research is that the Seasonal ARIMA method can serve as an effective decision-making tool for MSME (Micro, Small, and Medium Enterprise) operators, especially in planning inventory, production, and time-based marketing strategies. By understanding the seasonal sales patterns, businesses like KleponHolic can optimize performance during peak periods and anticipate downturns during low seasons through targeted promotions and operational efficiency. Furthermore, this study highlights the importance of regular and structured sales data recording as a foundation for implementing data-driven decision making.

From a theoretical perspective, this research contributes to the forecasting literature by providing empirical evidence of the applicability of Seasonal ARIMA in the context of culinary MSMEs in developing countries. Previous studies have predominantly focused on large-scale industries, manufacturing, or agriculture, whereas this study extends the scope of SARIMA application to small-scale food enterprises. The findings also strengthen the theoretical understanding that SARIMA remains competitive compared to newer methods (e.g., Prophet, LSTM) when dealing with strongly seasonal data, thereby enriching ongoing debates in forecasting methodology.

In terms of policy implications, the results emphasize the need for local governments and policymakers to promote the adoption of data-driven decision-making tools among MSMEs. This could be achieved through training programs on sales data recording and basic forecasting, or by integrating forecasting support systems into MSME development initiatives. Such policies would not only enhance the operational efficiency of small businesses but also strengthen the resilience of local economies against seasonal fluctuations in demand.

6. Conclusion

This study aims to model and forecast the sales of KleponHolic products using the Seasonal ARIMA method. Based on the analysis of monthly sales data from 2021 to 2024, the best-fit model

identified was Seasonal ARIMA (0,2,1)(0,1,0)[12], which proved effective in capturing both seasonal patterns and sales trends accurately. The forecasting results indicate sales fluctuations throughout 2025, with a tendency to decline at the beginning of the year and a significant surge during the middle and end of the year, particularly in December. The model also received validation from the business owner, who stated that the forecasted results align well with real-world conditions and are useful for business planning.

This study has several limitations that should be acknowledged. First, the scope of variables used in the analysis was limited, as the forecasting model relied solely on historical monthly sales figures without incorporating external factors such as promotional activities, consumer purchasing power, shifts in sales location, operating hours, or broader market conditions. These unobserved factors may have influenced fluctuations in sales but were not captured in the model. Future research should integrate additional explanatory variables or adopt SARIMAX and hybrid forecasting approaches to better account for these external influences.

Second, the dataset covered a relatively short period of four years (2021–2024). While this time frame was sufficient to identify seasonal patterns, it may not fully capture long-term structural changes in market demand or consumer behavior. Future studies could employ a longer dataset to improve model robustness and capture broader business cycles.

Third, this study applied only the Seasonal ARIMA model without conducting a systematic comparison against alternative forecasting methods such as Prophet, LSTM, or other machine learning approaches. Future research should include benchmarking to assess the relative strengths and weaknesses of SARIMA compared to newer approaches, especially in the context of MSMEs with seasonal demand.

Finally, model validation relied primarily on the input of a single business owner as the key informant. While this provided valuable practical insights, it limits the generalizability of the findings. Future research should involve multiple stakeholders or employ external validation methods, such as survey-based assessments or experimental implementation, to strengthen the credibility of the results.

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